SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	NoBugPls
Project / Solution Name	AiversePulse Enterprise
Name (Leader)	TANG YU SHAN
Email & Phone (Leader)	tangys2005@gmail.com & +60175908412
University Affiliation(s)	Universiti Teknologi Malaysia (UTM); Universiti Tunku Abdul Rahman (UTAR)
Live Prototype URL	https://sdc-hackathon-submission-no-bug-pls.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/yushan1101/SDC_Hackathon_Submission_NoBugPls

2. Architecture & Technology Stack

2.1 System Architecture Overview

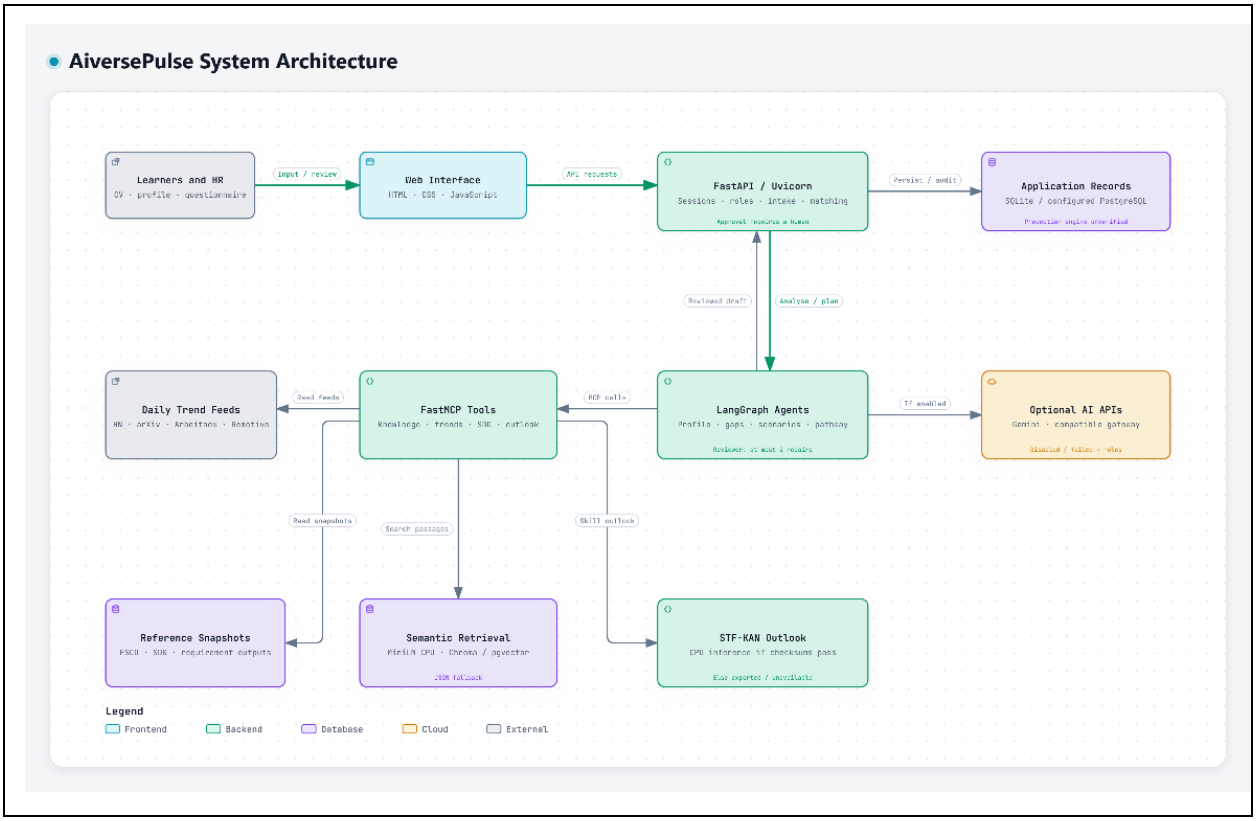
Describe how the components declared in this document connect and (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

AiversePulse serves an HTML/CSS/JavaScript interface through FastAPI/Uvicorn. Session authentication and role checks separate employee and HR workflows. CVs are parsed and screened; MiniLM embeddings normalize skills against ESCO. Editable profiles and questionnaires feed deterministic role matching, skill-gap analysis and learning plans. Matching combines skill coverage, alignment and optional embedding similarity; scores are not hiring probabilities.

LangGraph routes requests and coordinates profiler, knowledge, market, gap, pathway, coach and reviewer components, with bounded revisions. FastMCP exposes knowledge, trends and forecasting tools. Chroma retrieves attributed passages, with JSON fallback. SQLite persists accounts, plans, reports, recruitment, interviews, progress and audit events.

Gemini is optionally configured for routing and coaching; resume enrichment is opt-in. An OpenAI-compatible gateway serves per-task coaching, resume and voice models. MiniLM inference runs locally. STF-KAN supports checksum-gated CPU inference; without its serving artifacts it falls back to precomputed ESCO outlooks. Exported requirement-model predictions supply skill levels. Daily feeds combine Hacker News, arXiv, Arbeitnow and Remotive. ESCO and UN SDG snapshots support retrieval and mappings. The backend API runs on a dual-node VPS (main server at sdc.olgtx.com plus a worker node); Replit hosts the frontend site and development environment.



2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.13.0 local; supported >=3.11, <3.14	Application, data processing and model code. Replit specifies Python 3.11.	PSF license
HTML, CSS, JavaScript; SQLite 3.45.3	Browser interface and local relational storage. SQLite runs on the main VPS node.	Web standards; SQLite public domain
FastAPI 0.150.0 local; 0.156.4 locked; Uvicorn 0.54.0	HTTP API, sessions, uploads, streaming responses and web serving.	MIT; BSD-3-Clause
Pydantic 2.14.1; pydantic-settings 2.16.0 local / 2.16.1 locked	Typed request/state validation and environment configuration.	MIT
LangGraph 1.2.16; FastMCP 4.1.1 local / 4.1.3 locked	Agent routing, parallel analysis, review loop; callable knowledge and forecast tools.	MIT; Apache-2.0

Name & Version	Purpose in Project	License
sentence-transformers 6.1.0; ChromaDB 1.5.9	MiniLM embeddings; local vector retrieval with attributed JSON fallback.	Apache-2.0
PyTorch 2.14.1 local; 2.14.1+cpu locked	Embedding inference, requirement model and STF-KAN training/serving.	Apache-2.0 plus LLVM exception, BSD-2/3, BSL-1.0, MIT components
Transformers 5.20.2; huggingface-hub 1.33.0; tokenizers 0.23.3; safetensors 0.8.0	Transitive embedding/model-loading dependencies in the local environment.	Apache-2.0
NumPy 2.5.4; pandas 3.0.6; scikit-learn 1.9.1	Numerical arrays, data preparation, metrics and model experiments.	BSD-3-Clause; NumPy bundled components also 0BSD, MIT, Zlib, CC0
HTTPX 0.28.1; google-genai 2.31.0 local / 2.32.0 locked	Feed/payment/gateway HTTP requests; optional Gemini SDK.	BSD-3-Clause; Apache-2.0
pdfplumber 0.11.10; pikepdf 10.17.0; python-docx 1.2.0	PDF/DOCX extraction and document integrity checks.	MIT; MPL-2.0; MIT
python-multipart 0.0.32; Pillow 12.3.0	Multipart uploads and slider-challenge images.	Apache-2.0; MIT-CMU
Streamlit 1.66.0	Alternative UI module; the Replit launch command serves FastAPI.	Apache-2.0
pytest 9.1.1; Playwright (unversioned optional import)	Automated tests; optional browser-demo script imports Playwright.	MIT; Apache-2.0
pypdf 6.21.1; PyMuPDF 2.0	Declared/installed dependencies; no direct import found in audited application source.	BSD-3-Clause; PyMuPDF AGPL-3.0 or commercial license
XGBoost >=3.0 (declared range)	Optional offline rising-signal experiment.	Apache-2.0
RunPod SDK (unpinned); pytorch/pytorch:2.4.1-cuda12.1-cudnn9-runtime base image	GPU worker image for RunPod Serverless training and batch embedding jobs.	RunPod SDK MIT; PyTorch image BSD-3-Clause

Name & Version	Purpose in Project	License
uv (version not recorded); setuptools 84.2.0; wheel (build dependency)	Dependency locking, Replit launch and Python packaging.	uv MIT/Apache-2.0; setuptools MIT; wheel MIT
Node.js 20 (Replit config); Docker/Compose; nginx:1.27-alpine (example)	Development/build utilities; Docker and nginx examples for container and separate-frontend deployment.	Node MIT plus bundled terms; Docker/nginx image terms follow upstream distributions
Three.js r180; GSAP and ScrollTrigger 3.12.5	Vendored browser libraries for virtual interview/office graphics and UI animation.	Three.js MIT; GSAP files state the Standard License or Club GSAP agreement; exact applicable grant not recorded
Tailwind CSS (unpinned design-export CDN); Google Fonts (unversioned)	Tailwind in design exports; runtime EB Garamond, Geist and Geist Mono. Design exports also reference Domine, IBM Plex Sans, Public Sans and Material Symbols.	Tailwind MIT; EB Garamond OFL-1.1; other font/icon assets follow publisher terms
cryptography (Fernet); Node.js built-in test runner	Encryption of stored hiring-plugin tokens; frontend review-queue UI tests.	Apache-2.0 / BSD-3-Clause; Node.js MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API / Google (google-genai)	Optional router/coach and interview text; opt-in redacted CV enrichment. Local model setting: gemini-3.1-flash-lite. API-key authentication.	Paid per token (US$0.25 in / US$1.50 out per 1M tokens)
OpenAI-compatible gateway (configurable adapter)	Optional /v1/chat/completions via HTTPX and Bearer authentication. Configured per agent: coach grok-4.7, resume screening gpt-5.6-luna, voice input mimo-v2.6-flash.	Supports an optional OpenAI-compatible API gateway. The gateway provider, deployed endpoint, runtime configuration, actual API usage, account tier, and billing arrangement have not yet been verified and require confirmation by the team.
HN Search / Algolia: hn.algolia.com/api/v1/search	Public technology-news counts, titles and links for daily trend signals; no key in adapter.	Public unauthenticated access
arXiv: export.arxiv.org/api/query	Research metadata and counts for trend signals; adapter spaces requests.	Public unauthenticated access
Arbeitnow: www.arbeitnow.com/api/job-board-api	Job postings and skill-demand signals; no key in adapter.	Public unauthenticated access
Remotive: remotive.com/api/remote-jobs	Remote-job postings and trend signals. Attribution, linking and display restrictions apply.	Public API; terms apply
European Commission ESCO: ec.europa.eu/esco/api	Build-time occupation/skill snapshot download; runtime reads local derivatives.	Public access

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
UN Statistics Division: unstats.un.org/SDGAPI/v1/sdg/Goal/List	Build-time goals/targets download for local SDG mapping.	Public access; reuse terms to verify
Hugging Face Hub / huggingface.co	Pinned MiniLM download/cache and historical job-dataset acquisition; no remote inference required.	Free public assets
Stripe API / Stripe	Pro and Team plan checkout (MYR) via payment intents and a billing webhook; payments are off when keys are empty.	Test mode; live use is per-transaction paid
Hiring plugins: LinkedIn API, Facebook Graph API v23.0, X API (OAuth); Telegram, Discord and Slack	HR connects each platform once; a posting agent shares every new job opening. Stored platform tokens are Fernet-encrypted.	Free developer access through team-registered apps
SMTP email (configurable provider)	Sign-up verification, password reset and HR invite emails; disabled when no SMTP host is set.	Depends on the configured provider

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development environment and frontend site; workflow runs uv run uvicorn on port 5000, exposed as port 80. Frontend at sdc-hackathon-submission-no-bug-pls.replit.app calls the VPS backend.	SDC sponsor resource; plan and region to be confirmed by the team
RunPod	One NVIDIA A40 GPU Pod used for a standalone pretrained MiniLM skill-embedding inference experiment. No STF-KAN or requirement-model training; no production integration.	SDC sponsor resource; US$25 project cap; region to be confirmed by the team
GitHub	Source repository, version control and GitHub Actions CI on every push.	Free; public repository

Service & Provider	Purpose in Project	Plan / Region
Google Gemini; optional gateway, Stripe and SMTP providers	Conditional remote AI, payment and email services; no managed authentication provider identified.	Pay-as-you-go; Stripe in test mode; plans and regions to be confirmed by the team
Local application filesystem / SQLite / Chroma	Database, vector index, snapshots and model artifacts on the main VPS node. Docker/Compose deployment examples exist. No managed database service is used.	Main VPS node; backups and region to be confirmed by the team
Dual-node VPS (main + worker) at sdc.olgtx.com; Cloudflare Workers (MCP balancer)	Backend API host: main server at sdc.olgtx.com plus a worker node for heavy tools; optional Cloudflare Worker at mcp.olgtx.com balances MCP tool calls across workers.	Plan and region to be confirmed by the team

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
all-MiniLM-L6-v2; revision 1110a243fdf4706b3f48f1d95db1a4f5529b4d41	sentence-transformers / Hugging Face	Pretrained local embeddings: skills, role similarity, retrieval, SDG mapping and model features. Not fine-tuned.	Apache-2.0
gemini-3.1-flash-lite	Google Gemini	Pretrained API model; configured for router/coach and interviews; CV enrichment opt-in.	Gemini API Terms of Service
grok-4.7; gpt-5.6-luna; mimo-v2.6-flash	Generic OpenAI-compatible adapter	Pretrained API models via one OpenAI-compatible gateway: coaching, resume skill proposals and voice transcription; per-task model chosen by admin/HR.	Each model provider's API terms apply
STF-KAN, ensemble:stf_kanx3; export 2026-10-20	Team implementation; O*NET release-history inputs	Self-trained model outputs in ESCO snapshot. Checksum-gated CPU serving implemented; falls back to precomputed outlooks when artifacts are absent.	Team code Apache-2.0; O*NET attribution/CC BY terms
ESCO requirement MLP; model card 2026-10-20	Team implementation; MiniLM + O*NET 31.0	Self-trained importance/level regression on 140,381 rating pairs; runtime reads exported ESCO predictions.	Team code Apache-2.0; MiniLM Apache-2.0; O*NET CC BY 4.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ESCO v1.2 occupations and skills	European Commission; esco.ec.europa.eu	Public; runtime taxonomy and attributed knowledge	EC reuse policy; attribution	N
O*NET 31.0 ratings; releases 25.0–31.0	USDOL/ETA; onetcenter.org/database.html	Public; model training. History also needed for optional live forecast inference.	CC BY 4.0; preserve notices	N
data_jobs 2023; recorded rev ae47402	huggingface.co/datasets/Ashy24/data_jobs	Public mirror; offline forecasting experiment/aggregates	Mirror card says Apache-2.0; upstream rights and ad-content permission are not established	Y: possible ad contacts
UN SDGs: 17 goals, 169 targets	unstats.un.org/SDGAPI; sdgs.un.org/goals	Public; semantic mappings	UN API is public; reuse follows UN terms and is not assumed public domain	N
Hacker News news metadata	hn.algolia.com/api	Public feed; live trend counts and titles/links	Source/API terms; no blanket open license	Y: author metadata possible
arXiv paper metadata	info.arxiv.org/help/api	Public feed; research counts and metadata	arXiv API/content terms	Y: author names possible
Arbeitnow and Remotive postings	arbeitnow.com/api/job-board-api; remotive.com/api/remote-jobs	Public feeds; live job/trend evidence	Provider terms; Remotive linking/display restrictions	Y: contacts possible

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Official learning links and knowledge_passages.json	src/mcp_server/ingest.py; ESCO and official publisher URLs	Public/derived; attributed retrieval and learning references	Publisher terms; learning resources link-only	N intended
ESCO requirement/outlook predictions and SDG mappings	src/data/snapshots/; team model outputs	Derived; scores and scenario ranking signals	Team code Apache-2.0; source-data terms retained	N
Sample profiles, team CSV, sample CVs and test fixtures	src/data/samples/; tests/; seed_demo_users.py	Demo examples; not verified as deployed data	Repository labels some as samples; complete provenance/ownership not established	No personal data identified in inspected sample names/content; confirm before demo
User-submitted profiles, CV text, accounts and interview/recruitment records	Application forms and local SQLite	Private/self-collected; operational use, not declared training data	Application records may contain personal information; demo use requires public/synthetic inputs	Y

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI Codex (this declaration session; model/version not shown)	Coding assistance, UI modifications, debugging, technical explanations, documentation and declaration review. Repository audit and drafting this disclosure. This does not establish earlier application-development use.	Moderate for this declaration only; whole-project extent not inferable
Replit Agent/AI (version not recorded)	Replit workflow metadata says author="agent"; this is configuration metadata, not proof of team AI-assisted development.	Moderate; Assistance with application setup, development configuration
Google Stitch (version not recorded)	Stitch prompt and exported UI design files are present in the repository; their author and use in production code are not established.	Design artifacts present; integration extent unknown
No other named coding/research assistant found in reviewed files	Repository scan found no further tool identity. External team use is not represented by code.	No extent can be attributed from repository evidence
No image-generation tool identified in reviewed files	Asset provenance is not recorded; existing images alone do not establish generated content.	No extent can be attributed from repository evidence

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Team members reviewed AI-generated code and content before committing it. Every push runs the automated suite (74 pytest tests plus Node UI tests) in GitHub Actions CI. Model metrics come from held-out evaluation logged in docs/experiment_log.md. In the product, agent outputs pass a reviewer step, and a named person must approve every plan and hiring decision.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Development platform, Secrets and frontend hosting	Team development; API keys in Replit Secrets; .replit workflow serves the frontend site, which calls the VPS backend.
RunPod	NVIDIA A40 GPU Pod; PyTorch CUDA	Used a pretrained MiniLM model to generate embeddings for six example career skill descriptions on a RunPod GPU. No model training or fine-tuning was performed, and the experiment was not connected to the deployed website.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit and RunPod were used, but neither is essential to core functionality. The core FastAPI application, SQLite/Chroma storage, local embeddings and exported model outputs can run on another compatible host. The backend already runs on a VPS; the Replit frontend can move to any static host. The optional RunPod worker is not required by the request path. Remote LLM features have deterministic fallbacks, although removing them reduces narrative and interview-generation capabilities.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application stores account/contact details, profile skills and questionnaires, plans/reports, progress, interview answers, recruitment records and audit events in SQLite. Uploaded CVs are parsed; sanitized Markdown can be retained in applications. Regex redaction is imperfect. Optional LLM calls send selected context; resume enrichment sends redacted text when enabled. Stripe and SMTP receive payment/email data when configured.

Controls include salted scrypt passwords, hashed session tokens, role/ownership checks, upload limits, document screening and cookie/origin protections. Session cookies default to SameSite=Strict, and stored hiring-plugin tokens are Fernet-encrypted; the database itself is not encrypted at rest.

Self-service deletion removes specified account/workflow records and pseudonymizes some audit/payment references; it does not prove erasure of every HR/application record or backup. No universal automatic retention period is implemented. SMTP-less development mode logs OTP codes/links. Deployment retention and provider processing terms are not in repository configuration. Demonstration inputs should be limited to public or synthetic data.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

LLMs can hallucinate, misread context and produce biased advice. Implemented checks validate structured output, allowed citations and quoted CV evidence; the reviewer has a bounded revision loop. MiniLM similarity and rule-based extraction can miss skills or infer incorrect matches; users can edit profiles. Matching scores are heuristic rankings, not calibrated employability probabilities. ESCO/O*NET coverage and transferred requirement levels may not represent local jobs. STF-KAN forecasts historical release ratings, not current vacancies; its recorded overall error does not beat the last-value baseline. Direction gates and precomputed/unavailable fallbacks limit unsupported claims. SDG links are semantic associations.

Public feeds can be stale, biased or rate-limited; snapshots/cache fallback preserve partial operation. Optional LLM providers add availability, cost and privacy risks; regex redaction may miss identifiers.

Human review remains necessary: career and interview recommendations support decisions and must not be treated as final hiring decisions.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



[Signature]

24/10/2026

[Date]